

17ELMO1

SomeonecoolLovesMaths

June 2, 2026

Problem

Let $a_1, a_2, \dots, a_n$ be positive integers with product P , where n is an odd positive integer. Prove that

$$\gcd(a_1^n + P, a_2^n + P, \dots, a_n^n + P) \leq 2 \gcd(a_1, \dots, a_n)^n.$$

Solution. Let $\gcd(a_1, \dots, a_n) = d$. Define $b_i = \frac{a_i}{d}$ and $Q = \frac{P}{d^n}$. Observe that $\gcd(b_1, \dots, b_n) = 1$.

So it is enough to prove that $\gcd(b_1^n + Q, b_2^n + Q, \dots, b_n^n + Q) = g \leq 2$.

We have $b_i^n \equiv -Q \pmod{g}$ for all $1 \leq i \leq n$. So, $\prod_{i=1}^n b_i^n = Q^n \equiv (-Q)^n = -Q^n \pmod{g}$. Say $g \geq 3$. We have $g \mid 2Q^n$.

Since $g \geq 3$, there is a prime p such that $p \mid Q^n \implies p \mid Q$. Since $g \mid b_i^n + Q \implies p \mid b_i^n \implies p \mid b_i$ for all $1 \leq i \leq n$. This of course contradicts the fact that $\gcd(b_1, \dots, b_n) = 1$.

So, $g \leq 2$.